Name: Hasnain Qazi
Sap Id: 82739
Program: BS(AI)
Subject: Data Structures

Assignment #2

Q1- Singly Linked List - Patient Registration

A singly Linked List is a linear data structure in which every node stores patient information and one pointer named next. The next pointer contains the address of the next node. The last node points to NULL. Patient records contains: Patient ID, Patient Name, Age, Disease and next pointer.

Required Operations

- Add a new patient: create a new node and attach it at the end of the list.
- Search by ID: traverse from head and compare each patient ID.
- Remove by ID: locate the patient, reconnect the links and delete that node.
- Display all patients: start from head and print each node until NULL

Algorithms

A. Add Patient

1- Create a new patient node using dynamic memory
2- Store ID, name, age and disease in the node.
3- Set newNode -> next = NULL.
4- If head is NULL, set head = newNode.
5- Otherwise move to the last node and set last->next=newNode

B. Search Patient by ID

1- Start current from head.
2- Compare current patient ID with the required ID.
3- If IDs match, display the patient and stop.
4- Otherwise move current = current -> next
5- If current becomes NULL, report patient not found.

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C. Remove Patient by ID

1- If the list is empty, stop.
2- If the head node has the required ID, move head -> next and delete the old head.
3- Otherwise find the node whose next node contains the required ID.
4- Connect current -> next to the node after the deleted node.
5- Delete the removed node from memory.

D. Display Patients

1- Start current from head.
2- Display ID, name, age and disease.
3- Move current to current -> next
4- Repeat until current becomes NULL.

Diagram

Head

[101|Ali|25|Fever] -> [102|Ahmed|34|Flu] -> [103|Sara|28|Cough] -> NULL

Q2. Doubly Linked List - Patient History

A doubly linked list is a linked list in which each node contains two links: prev and next, so patient history can be viewed forward and backward.

Operations

- Add treated patient
- Display forward
- Display reverse

Add: Create node. If empty, head = newNode. Otherwise connect it after tail and update tail.

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Forward: Start at head and follow next until NULL
Reverse: Start at tail and follow prev until NULL

NULL <- [101|Ali] <=> [102|Ahmed] <=> [103|Sara] -> NULL
Head Tail

Q3: Circular Linked List - Appointment Management

In a circular linked list, the last node points back to head instead of NULL. It is useful for repeating appointments.

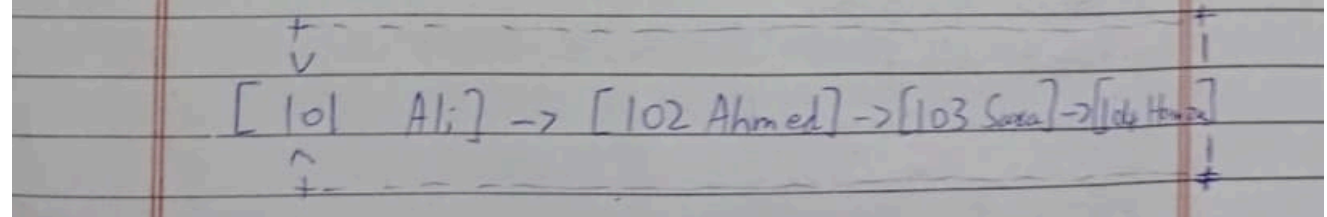
Operations:

Add appointment,
Display once,
Move to next appointment

Add: Create node. If empty, head = newNode. and newNode -> next = head. Otherwise attach after last and set newNode -> next = head.

Display: Start at head and prints nodes until the pointer reaches head again.

Next: CurrentAppointment = CurrentAppointment -> next. After the last patient it returns to head.



Q4: Stack ADT - Emergency Management

A stack follows LIFO (Last In, First Out). The

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latest emergency patient added is removed first.

TOP -> [Sara]
[Ahmed]
[Ali]

Push:
Create node -> newNode -> next = top -> top = newNode

PoP: If not empty, save top -> move top to top -> next -> delete old top.

Peek: Display the patient at top without deleting it.

Display: Start at top and follow next until NULL.

Q5 Menu Driven Program and Demonstration

The program combines all four data structures in one hospital management system.

Module	Data Structure
Patient Registration	Singly Linked List
Patient History	Doubly Linked List
Appointment Management	Circular Linked List
Emergency Management	Stack ADT

Main Menu
1- Patient Registration
2- Patient History
3- Appointment Management
4- Emergency Management
5- Exit

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Five Patient Records

ID	Name	Age	Disease
101	Ali	25	Fever
102	Ahmed	34	Flu
103	Sara	28	Cough
104	Hamza	42	Diabetes
105	Ayesha	31	Asthma

Short Demonstration / Sample Output

Registered Patients:
101 Ali, 25, Fever
102 Ahmed, 34, Flu
103 Sara, 28, Cough
104 Hamza, 42, Diabetes
105 Ayesha, 31, Asthma

Search ID 103 -> Patient Found : Sara
Remove ID 102 -> Patient removed successfully

History Forward: 101 Ali -> 102 Ahmed -> 103 Sara
History Reverse: 103 Sara -> 102 Ahmed -> 101 Ali

Appointments: 101 Ali -> 102 Ahmed -> 103 Sara -> 104 Hamza -> back to 101
Next Appointment: 102 Ahmed

Emergency Stack (Top to Bottom):
105 Ayesha
103 Sara

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101 Ali
Peek: 105 Ayesha
Pop: 105 Ayesha removed.